

A Branch-Safe Spectral and Scattering Atlas for a Side-Coupled Fano–Anderson Impurity

Route-A source-local certificate HCS-C345

3 September 2026

Abstract

We solve the infinite nearest-neighbour chain with one discrete state side-coupled at the origin. A physical-sheet Schur function gives exactly two simple bound states for every nonzero coupling, one on each side of the free band. Squaring the secular equation can create extra real roots; an explicit sign filter prevents their promotion to eigenvalues. Reflection parity and a cyclic eventually-free Jacobi representation yield the complete spectral type, band multiplicity, impurity density, residue weights, and mass normalization. Stationary matching gives exact unitary reflection and transmission laws and the if-and-only-if location of the Fano zero. The decoupled, zero-hopping, band-edge, coupling-sign, and zero-coupling-limit faces are closed separately. Exact finite receipts test branch rejection but do not replace the infinite-volume proof. Natural quantization survives only at A4; Route A is rejected and Route B remains locked.

1 Frozen Hamiltonian and physical sheet

Let

$$\mathcal{H} = \ell^2(\mathbb{Z}) \oplus \mathbb{C}|d\rangle, \quad J > 0, \quad \varepsilon \in \mathbb{R}, \quad g \in \mathbb{R},$$

and define

$$(Hu)_n = J(u_{n+1} + u_{n-1}) + gu_d \delta_{n0}, \quad (1)$$

$$(Hu)_d = \varepsilon u_d + gu_0. \quad (2)$$

This is a bounded self-adjoint operator on all of $\mathcal{H}$. Write H_0 for the free chain. Fourier transform makes H_0 multiplication by $2J \cos k$, so

$$\sigma(H_0) = \sigma_{\text{ac}}(H_0) = [-2J, 2J]$$

with multiplicity two in the open band.

Fix the analytic branch

$$S(z) = \sqrt{z^2 - 4J^2}, \quad S(z)/z \rightarrow 1 \quad (z \rightarrow \infty).$$

Thus $S(E) > 0$ for $E > 2J$ and $S(E) < 0$ for $E < -2J$. A contour integral, with the normalization checked at infinity, gives

$$m(z) = \langle 0, (z - H_0)^{-1} 0 \rangle = \frac{1}{S(z)}. \quad (3)$$

Block inversion at $|d\rangle$ now yields

$$G_{dd}(z) = \langle d, (z - H)^{-1} d \rangle = \frac{1}{D(z)}, \quad D(z) = z - \varepsilon - \frac{g^2}{S(z)}. \quad (4)$$

Theorem 1 (Two physical-sheet bound states). *If $g \neq 0$, H has exactly two exterior eigenvalues. They are simple and are the unique solutions*

$$E_+ - \varepsilon - \frac{g^2}{\sqrt{E_+^2 - 4J^2}} = 0, \quad E_+ > \max(2J, \varepsilon), \quad (5)$$

$$E_- - \varepsilon + \frac{g^2}{\sqrt{E_-^2 - 4J^2}} = 0, \quad E_- < \min(-2J, \varepsilon). \quad (6)$$

Squaring produces

$$(E - \varepsilon)^2 (E^2 - 4J^2) - g^4 = 0, \quad (7)$$

but a real root of (7) is an eigenvalue only when it also satisfies the corresponding inequality in (5) or (6).

Proof. For $E > 2J$ put

$$D_+(E) = E - \varepsilon - \frac{g^2}{\sqrt{E^2 - 4J^2}}.$$

It tends to $-\infty$ at $2J+$ and to $+\infty$ at infinity, while

$$D'_+(E) = 1 + \frac{g^2 E}{(E^2 - 4J^2)^{3/2}} > 0.$$

Hence it has exactly one simple zero. The equation itself forces $E_+ > \varepsilon$. Below the band use

$$D_-(E) = E - \varepsilon + \frac{g^2}{\sqrt{E^2 - 4J^2}}.$$

It tends to $-\infty$ at minus infinity and to $+\infty$ at $-2J-$, and

$$D'_-(E) = 1 - \frac{g^2 E}{(E^2 - 4J^2)^{3/2}} > 0.$$

The unique zero is simple and satisfies $E_- < \varepsilon$.

At an exterior zero, equations (1)–(2) give the square-summable vector $u_{\mathbb{Z}} = gu_d(E - H_0)^{-1}|0\rangle$. Conversely, an exterior eigenvector cannot have $u_d = 0$, because H_0 has no exterior eigenvalue; its impurity equation therefore gives $D(E) = 0$. This establishes the pole/eigenvalue equivalence. Squaring loses the sign of $S(E)$ and the sign of $E - \varepsilon$, which proves the stated filter. $\square$

The phrase *branch-safe Schur owner* marks the baseline manuscript: the quartic is an algebraic candidate ledger, never an unfiltered spectrum.

2 Full spectral measure

Let reflection send u_n to u_{-n} . The odd and even subspaces reduce H . On the odd subspace, $u_0 = u_d = 0$ and H is the free Dirichlet half-line Jacobi matrix. It has simple purely absolutely continuous spectrum on $[-2J, 2J]$.

On the even-plus-impurity subspace, use the ordered basis

$$|d\rangle, |0\rangle, \quad |e_n\rangle = \frac{|n\rangle + |-n\rangle}{\sqrt{2}} \quad (n \geq 1).$$

The Jacobi diagonal is $\varepsilon, 0, 0, \dots$ and its successive off-diagonals are $g, \sqrt{2}J, J, J, \dots$. If $g \neq 0$, $|d\rangle$ is cyclic. Its scalar resolvent is (4), so that scalar measure therefore determines the entire even spectral type.

Let μ_d be that scalar spectral measure. Our convention is

$$G_{dd}(z) = \int_{\mathbb{R}} \frac{d\mu_d(x)}{z - x}.$$

Thus G_{dd} is an anti-Herglotz Cauchy transform on the upper half-plane; $M_{dd} = -G_{dd} = \langle d, (H - z)^{-1}d \rangle$ is the standard Herglotz function. In particular the density has the minus sign below.

For $|E| < 2J$, set $s(E) = \sqrt{4J^2 - E^2}$. The upper boundary value is

$$G_{dd}(E + i0) = \frac{1}{E - \varepsilon + ig^2/s(E)}.$$

Stone–Stieltjes inversion gives the positive density

$$\rho_d(E) = -\frac{1}{\pi} \operatorname{Im} G_{dd}(E + i0) = \frac{g^2 s(E)}{\pi \{(E - \varepsilon)^2 s(E)^2 + g^4\}}. \quad (8)$$

Here is the measure-theoretic closure of this formula. Fix $K \Subset (-2J, 2J)$. The chosen square root has a continuous upper boundary value on K , and the denominator in the preceding display has nonzero imaginary part $g^2/s(E)$. Hence $G_{dd}(E + i\eta) \rightarrow G_{dd}(E + i0)$ uniformly on K . For every $\phi \in C_c(K)$, Stone's formula, followed by this uniform limit, gives

$$\int \phi d\mu_d = \lim_{\eta \downarrow 0} \int \phi(E) \left[-\frac{1}{\pi} \operatorname{Im} G_{dd}(E + i\eta) \right] dE = \int \phi(E) \rho_d(E) dE. \quad (9)$$

Compact exhaustion therefore identifies the full open-band measure, not only its absolutely continuous part.

On each real exterior, G_{dd} is real analytic except at the unique simple physical pole proved in Theorem 1. Formula (9) on pole-free exterior intervals gives zero measure there, and the two poles give precisely two atoms. Finally, for $E_0 = \pm 2J$,

$$G_{dd}(z) = \frac{S(z)}{(z - \varepsilon)S(z) - g^2} \longrightarrow 0, \quad \mu_d(\{E_0\}) = \lim_{\eta \downarrow 0} i\eta G_{dd}(E_0 + i\eta) = 0.$$

Any measure supported on the two leftover edge points would be atomic. Thus the density, the two exterior atoms, and the zero edge masses exhaust μ_d ; no singular continuous component or embedded atom remains.

At an exterior pole E_* , differentiation of (4) gives

$$w_* = \operatorname{Res}_{z=E_*} G_{dd}(z) = \frac{1}{1 - g^2 m'(E_*)}. \quad (10)$$

On both physical exteriors $m'(E_*) < 0$, hence $0 < w_* < 1$. Moreover

$$G_{dd}(z) = \frac{1}{z} + \frac{\varepsilon}{z^2} + \frac{\varepsilon^2 + g^2}{z^3} + \frac{\varepsilon^3 + 2\varepsilon g^2}{z^4} + O(z^{-5}). \quad (11)$$

The coefficient of z^{-1} in the Cauchy transform is the total mass, so

$$\int_{-2J}^{2J} \rho_d(E) dE + w_- + w_+ = 1. \quad (12)$$

Theorem 2 (Complete spectral type). *For $J > 0$ and $g \neq 0$,*

$$\sigma(H) = [-2J, 2J] \cup \{E_-, E_+\}, \quad \sigma_{\text{sc}}(H) = \emptyset.$$

The band is purely absolutely continuous of multiplicity two almost everywhere; $E_{\pm}$ are simple atoms with weights (10) in the impurity measure. Neither band edge is an eigenvalue.

Proof. The odd component contributes one simple absolutely continuous copy of the band. Cyclicity, the explicit scalar measure above, and Theorem 1 give one further simple absolutely continuous copy and exactly the two simple atoms in the even component. At $E = \pm 2J$, the free recurrence has affine rather than decaying half-line solutions. An ℓ^2 solution must vanish on both tails and then equations (1)–(2) force all remaining coordinates to vanish. Thus there is no edge eigenvector. $\square$

3 Stationary scattering and the Fano zero

Let $E = 2J \cos k$ with $0 < k < \pi$. Away from $E = \varepsilon$, eliminating u_d gives the on-shell origin potential

$$V(E) = \frac{g^2}{E - \varepsilon}.$$

Matching an incident wave, its reflection, and its transmission at the origin gives, in a consistent phase convention,

$$t(E) = \frac{2iJ \sin k}{2iJ \sin k + V(E)}, \quad r(E) = \frac{-V(E)}{2iJ \sin k + V(E)}. \quad (13)$$

Therefore

$$T(E) = \frac{4J^2 \sin^2 k (E - \varepsilon)^2}{g^4 + 4J^2 \sin^2 k (E - \varepsilon)^2}, \quad (14)$$

$$R(E) = \frac{g^4}{g^4 + 4J^2 \sin^2 k (E - \varepsilon)^2}, \quad T(E) + R(E) = 1. \quad (15)$$

At $E = \varepsilon$, elimination is invalid, but the original impurity equation gives $gu_0 = 0$. Thus $u_0 = 0$ and $T = 0$ directly. Consequently a continuum Fano zero occurs at $E = \varepsilon$ if and only if $|\varepsilon| < 2J$; equality is a closed channel edge, not an interior antiresonance.

The phrase *spectral scattering atlas owner* marks the first substantive revision: full spectral type, density, residues, mass, amplitudes, and antiresonance have been added to the branch-safe pole theorem.

4 Degenerate faces

If $g = 0$, the operator is the direct sum of the free chain and the isolated level ε . The latter is an embedded eigenvalue for $|\varepsilon| < 2J$, an eigenvalue at a band edge for equality, and an isolated eigenvalue outside the band. This face cannot be obtained by merely substituting $g = 0$ into formulas whose derivation divided by g .

Changing g to $-g$ is conjugation by the unitary that multiplies $|d\rangle$ by -1 . Thus the sign of the coupling changes phases in a chosen coordinate convention but not the spectrum, density, T , or R .

At $J = 0$, the origin and impurity form the two-dimensional block B_0 with entries $(B_0)_{00} = 0$, $(B_0)_{0d} = (B_0)_{d0} = g$, and $(B_0)_{dd} = \varepsilon$. Its eigenvalues are

$$E_{\pm}^{(0)} = \frac{\varepsilon \pm \sqrt{\varepsilon^2 + 4g^2}}{2},$$

while every other lattice coordinate belongs to an infinite-multiplicity zero eigenspace. This pure-point degeneration is not assigned the band multiplicity of Theorem 2.

The weak-coupling pole limit is nonuniform. If $|\varepsilon| < 2J$, the two poles approach the two band edges. If $\varepsilon > 2J$, the upper pole approaches ε and the lower pole approaches $-2J$; for $\varepsilon < -2J$ the reflected statement holds. At $\varepsilon = \pm 2J$, both descriptions meet at a square-root threshold. These limits do not replace the exact direct-sum classification at $g = 0$.

5 Exact evidence and collision boundary

The canonical receipt contains 126 signed-coupling Sturm rows, 315 rational scattering/density rows, 63 Fano-location rows, and 21 resolvent-moment rows, for 6,418 audited scalar leaves. Every Sturm row has one root in each physical interval. Sixteen rows have two additional real roots of the squared quartic, all recorded as branch-rejected. This directly exercises the central failure mode.

An independent checker rebuilds every row and evaluator field without importing the evidence generator. A separate SymPy lane checks the analytic identities, resolvent sign, edge limits, and root counts; two isolated directories reproduce identical evidence; repaired-hash mutations attack the six-field measure-proof lock, theorem fields, branch signs, Route-A fields, parsers, and every evaluator leaf. These are implementation receipts. The infinite-volume conclusions rest on the proofs above, never on a finite box or sampled convergence.

Fano [1] and Friedrichs [2] establish the discrete–continuum lineage; Miroshnichenko, Flach, and Kivshar [3] review Fano interference in discrete and nanoscale settings. They support provenance, not a priority claim for the present formulas or assembly.

The nearest workspace owners are C267 (uniform-field Wannier–Stark), C288 (continuum delta interaction), C308 (non-self-adjoint Hatano–Nelson), and C318 (dimerized SSH). None owns a side-coupled discrete level, two physical-sheet bound states, and the associated Fano zero. This is only a workspace collision statement.

6 Route-A decision and limitations

The phrase *quartic firewall and Route decision* marks the final revision. Under NO_BAD_EULER_OR_ROOT_NUMBER, the tuple is

(A0_FAIL, A1_FAIL, A2_FAIL, A3_FAIL, A4_NATURAL_QUANTIZATION).

The Hamiltonian, unitary group, and scattering problem make A4 natural, but there is no arithmetic origin, prime-power clock, arithmetic primitive-orbit ledger, target Euler product, target determinant, target functional equation, target counting law, Weil compression, or target-zero correspondence. Route A is therefore rejected and Route B remains locked. No target arithmetic local data, Euler factor, root number, automorphy, target divisor, Hilbert–Polya operator, or target-zero match is claimed.

AI use. A generative language model assisted proof organization, code scaffolding, and manuscript drafting. The explicit analytic derivation, independent checker, symbolic lane, hostile tests, and deterministic artifacts define the audit record.

References

- [1] U. Fano, *Effects of configuration interaction on intensities and phase shifts*, Physical Review 124 (1961), 1866–1878. DOI: [10.1103/PhysRev.124.1866](https://doi.org/10.1103/PhysRev.124.1866).
- [2] K. O. Friedrichs, *On the perturbation of continuous spectra*, Communications on Pure and Applied Mathematics 1 (1948), 361–406. DOI: [10.1002/cpa.3160010404](https://doi.org/10.1002/cpa.3160010404).
- [3] A. E. Miroshnichenko, S. Flach, and Y. S. Kivshar, *Fano resonances in nanoscale structures*, Reviews of Modern Physics 82 (2010), 2257–2298. DOI: [10.1103/RevModPhys.82.2257](https://doi.org/10.1103/RevModPhys.82.2257).